Huy Dinh Tran

huydinhtran@ku.edu — +1 (814) 826-8581 — LinkedIn — GitHub — Website — Google Scholar

EDUCATION

University of Kansas 08/2023 — 05/2028 (Expected)
Ph.D. in Computer Science
Advisor: Prof. Mohammad Alian (Cornell University)

University of California, Riverside 09/2021 — 03/2023
M.S. in Computer Engineering
Advisor: Prof. Daniel Wong

Pennsylvania State University 08/2017 — 05/2021
B.S. in Electrical Engineering

PUBLICATIONS

Amin Mamandipoor*, **Huy Dinh Tran***, Mohammad Alian, "SDT: Cutting Datacenter Tax Through Simultaneous Data-Delivery Threads," *IEEE CAL 2025*

* Equal contribution

EXPERIENCE

University of Kansas Lawrence, KS
Graduate Research Assistant 08/2023 — Present

- **Agentic LLM Inference Systems:** characterized host-CPU bottlenecks in multi-GPU agentic LLM serving (vLLM + OpenHands); cut the inference engine's core demand by 2× with user-level UMWAIT waits, and built a calibrated vLLM benchmark for gem5 to evaluate an asymmetric SMT core design.
- **Multithreading Architecture — Simultaneous Data-Delivery (SDD):** designed lightweight hardware threads that run alongside the main application thread to handle data delivery in server workloads; reduced area by 35% and power by 44%, while maintaining or exceeding throughput and tail latency performance.

Graduate Teaching Assistant 08/2024 — Present

- **Software Engineering II (EECS 581) and Senior Capstone (EECS 582):** supervised 7 senior undergraduate teams on design projects through weekly mentoring and technical guidance.

Futurewei Technologies San Jose, CA
Research Intern 08/2022 — 09/2022

- Simulated RISC-V CPUs using gem5 full-system simulation.
- Cross-compiled binaries of SPEC CPU 2017 benchmarks to RISC-V for measuring the performance of CPU designs.
- Integrated SimPoint to checkpoint regions of interest (ROIs), accelerating simulations while preserving representativeness.

PROJECTS

Simultaneous Multithreading (SMT) for gem5 2024 — 2025
Enabled SMT in gem5 O3CPU by restoring per-thread pipeline and interrupt handling, lifting the full-system single-thread restriction; implemented static and adaptive partitioning of microarchitectural resources across SMT threads.

Pipelined Execution for DPDK Userspace Networking Applications 2024
Implemented pipelined execution for DPDK-based userspace networking applications by decoupling NIC polling from application processing to enable parallelism; applied to DPDK l2fwd, Memcached and NGINX.

GPU Power Modeling with AccelWattch 2022 — 2023
Implemented a GPU power model using AccelWattch; profiled NVIDIA GTX 1050 Ti GPU to capture power usage and performance counters; simulated benchmarks in GPGPU-Sim to collect power model estimates.

SKILLS

Languages: C, C++, Python, Bash, CUDA, Verilog, MATLAB, LaTeX
AI/LLM Systems: OpenHands, Hugging Face Datasets, vLLM
Architecture & Systems: gem5, GPGPU-Sim, McPAT, DPDK, Linux perf, Intel VTune, Intel CAT
Infrastructure & Development: Linux, Docker, Git, GDB, QEMU, OpenMP